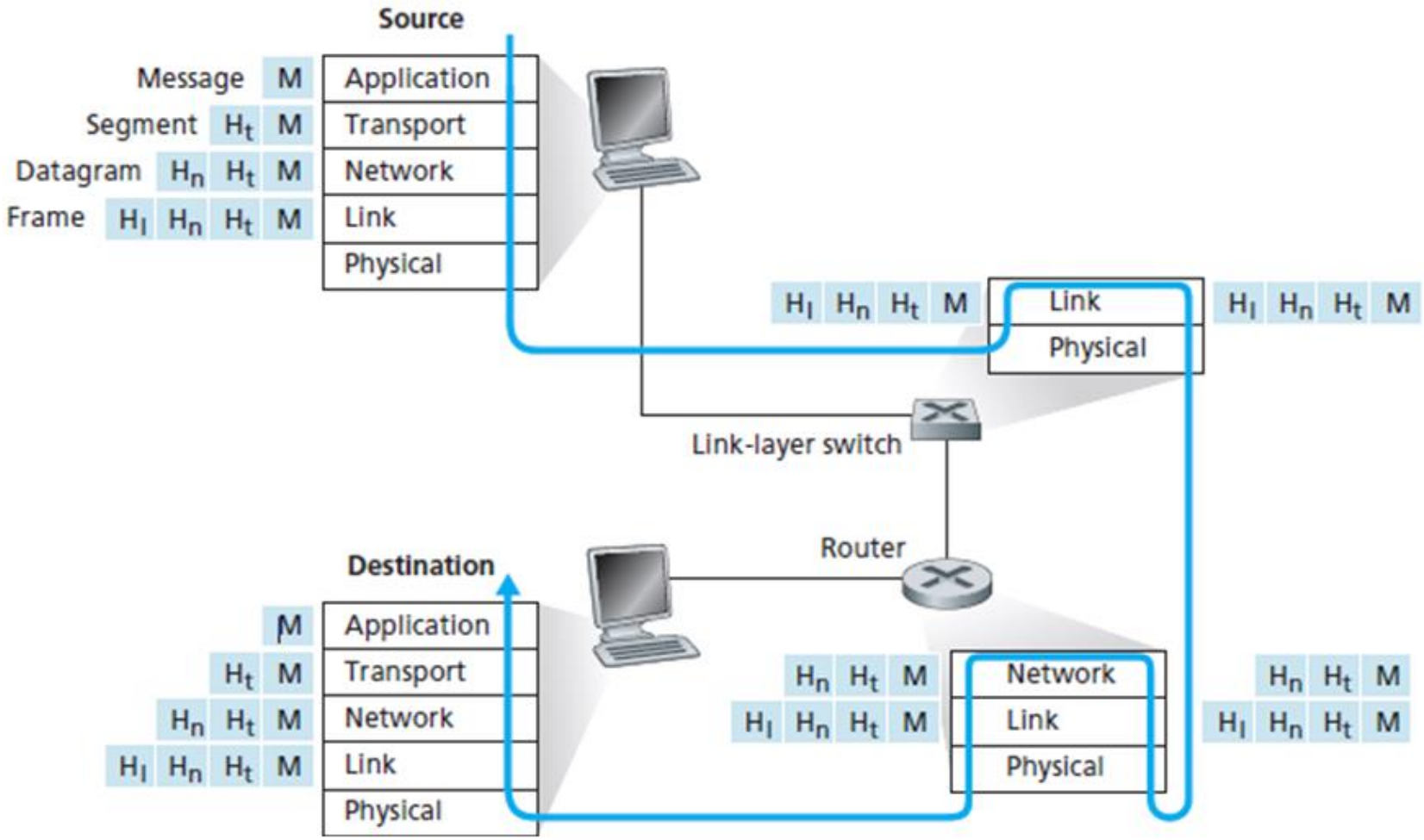
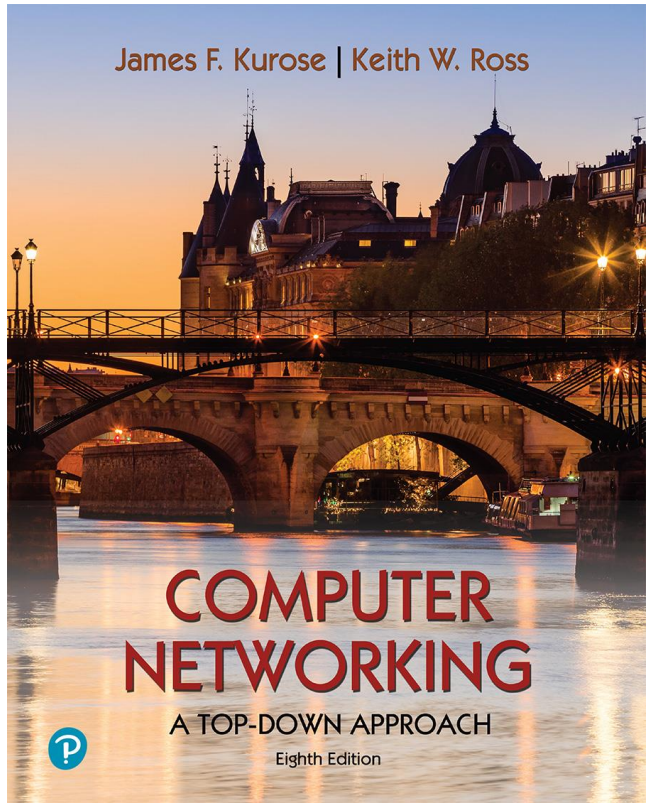
Chapter 4
Network Layer: Data Plane



Hosts, routers, and link-layer switches; each contains a different set of layers, reflecting their differences in functionality



Computer Networking: A Top-Down Approach
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

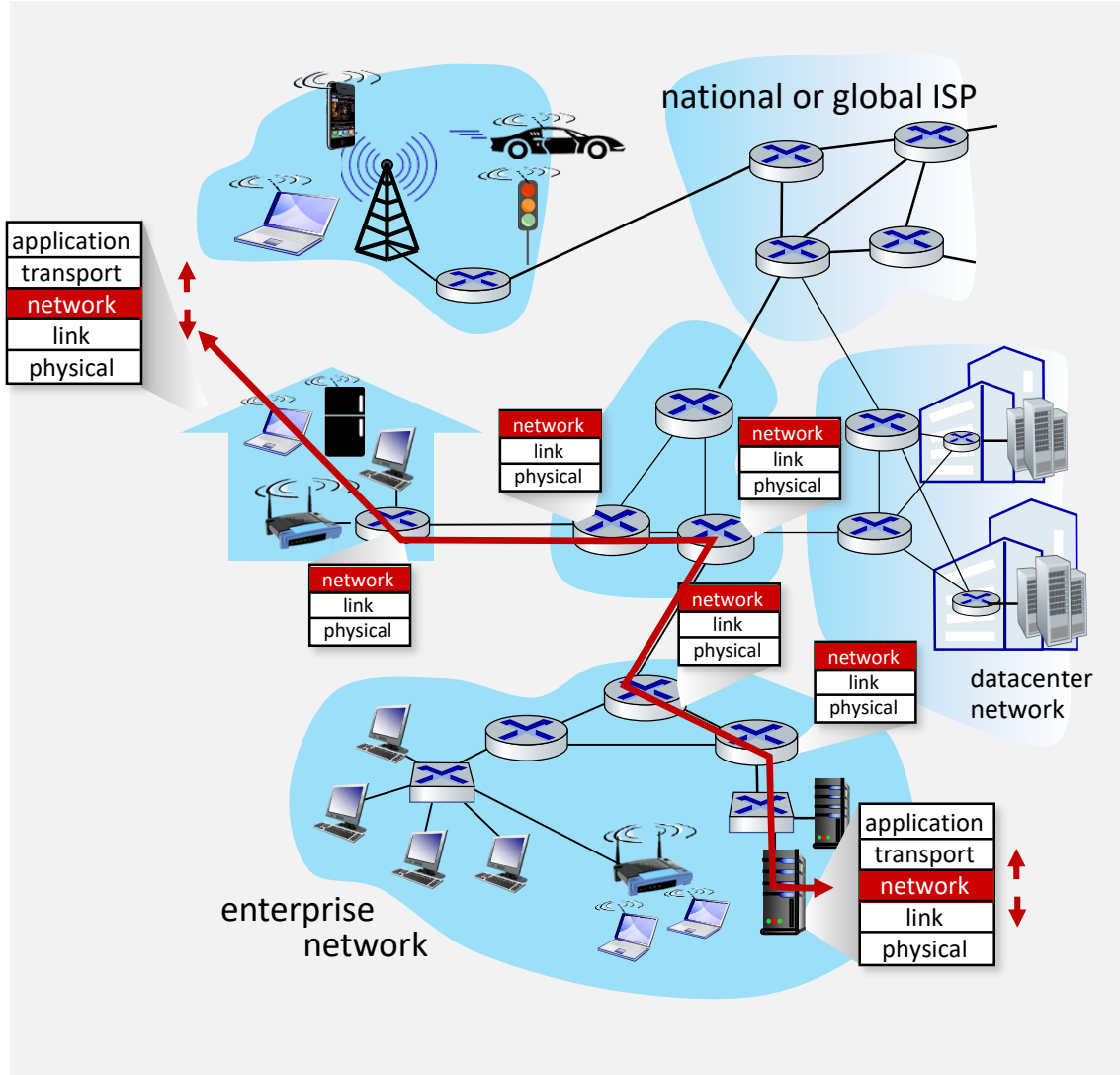
Introduction to Network layer

- The Internet's network layer routes a datagram through a series of routers between the source and destination.

- To move a packet from one node (host or router) to the next node in the route, the network layer relies on the services of the link layer.

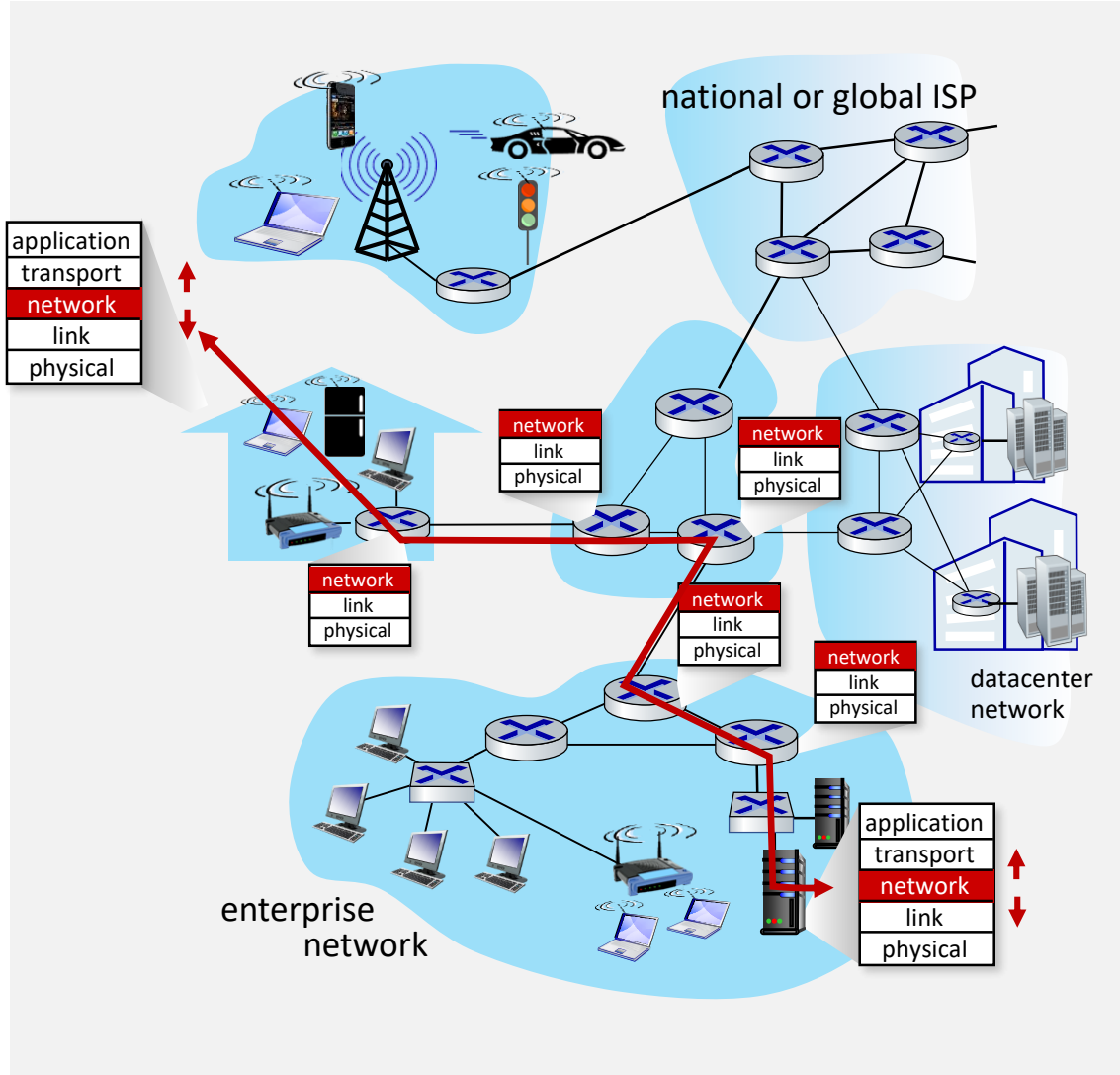
Network-layer services

- Transport segment from sending to receiving host
  - sender: encapsulates segments into datagrams, passes to link layer
  - receiver: delivers segments to transport layer protocol



Network-layer protocols

- Network layer protocols in every Internet device: hosts, routers
- Routers:
  - Examines header fields in all IP datagrams passing through it
  - Moves datagrams from input ports to output ports to transfer datagrams along end-end path



Introduction to Network layer

Network layer can be decomposed into two interacting parts, the **data plane** and the **control plane**.

Data plane:

- *local*, per-router function
- determines how datagram arriving on router input port is forwarded to router output port

Control plane:

- *network-wide* logic
- determines how datagram is routed among routers along end-end path from source host to destination host

- Assume a simple network with two hosts, H1 and H2 , and several routers on the path between H1 and H2.

- The primary data-plane role of each router is to forward datagrams from its input links to its output links.

- The primary role of the network control plane is to coordinate these local, per-router forwarding actions so that datagrams are ultimately transferred end-to-end, along paths of routers between source and destination hosts.

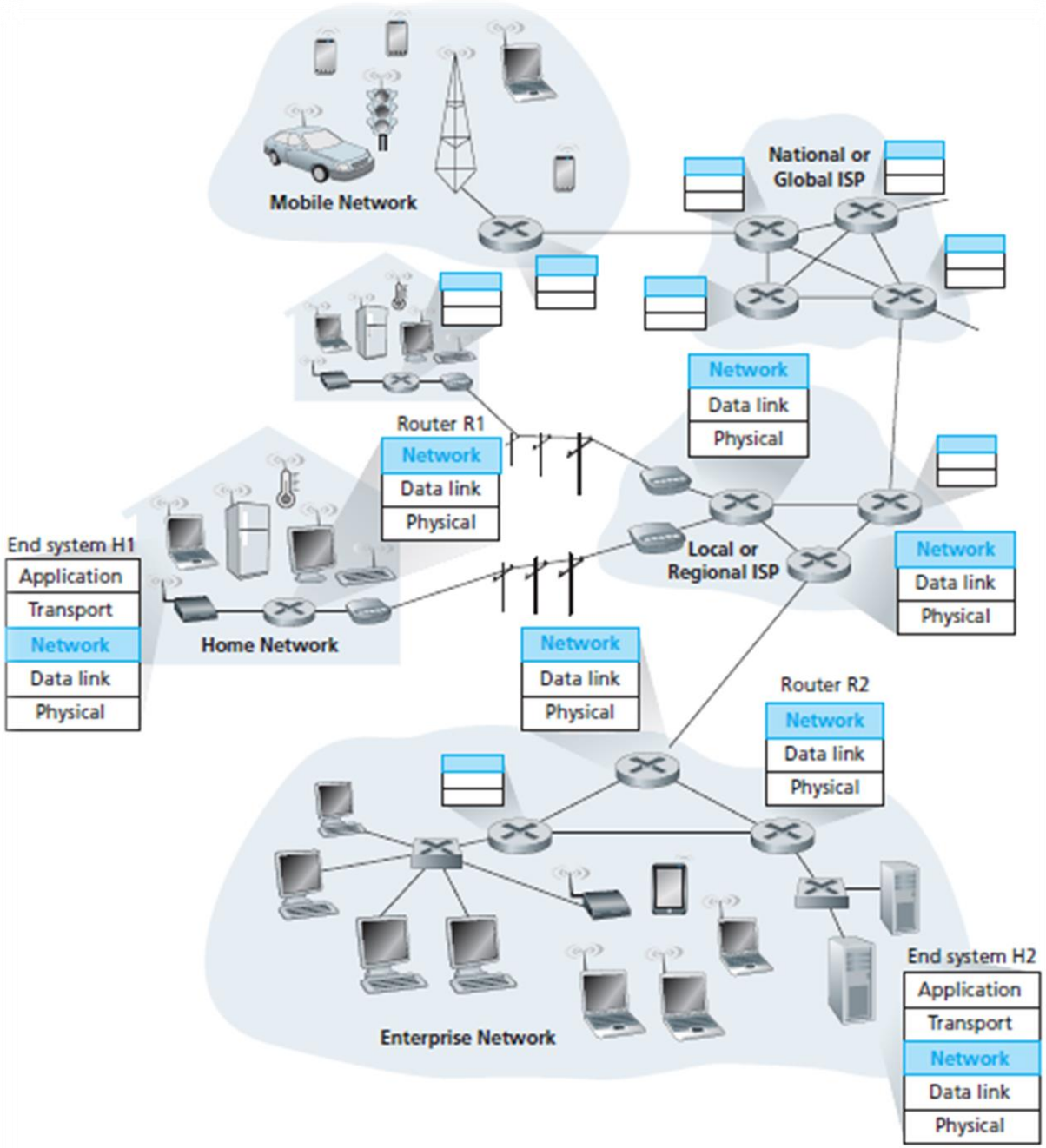


Figure 4.1 • The network layer

Primary Role of the Network Layer

- The primary role of the network layer is to move packets from a sending host to a receiving host. To do so, two important network-layer functions can be identified:

  - Forwarding
  - Routing

Forwarding

- When a packet arrives at a router's input link, the router must move the packet to the appropriate output link.
- For example, a packet arriving from Host H1 to Router R1 must be forwarded to the next router on a path to H2.
- Forwarding is but one function implemented in the data plane.

Forwarding

- **Forwarding** refers to the router-local action of transferring a packet from an input link interface to the appropriate output link interface.

- Forwarding takes place at very short timescales (typically a few nanoseconds), and thus is typically implemented in hardware.

Routing

- The network layer must determine the route or path taken by packets as they flow from a sender to a receiver.
- The algorithms that calculate these paths are referred to as **routing algorithms**.
- The routing algorithm determines the contents of the routers' forwarding tables.
- A routing algorithm would determine, for example, the path along which packets flow from host H1 to host H2.
- Routing is implemented in the control plane of the network layer.

Routing

- **Routing** refers to the network-wide process that determines the end-to-end paths that packets take from source to destination.

- Routing takes place on much longer timescales (typically seconds), and as we will see is often implemented in software.

Forwarding Table

- Forwarding is the key function performed by the data-plane functionality of the network layer.

- A key element in every network router is its **forwarding table**.
- A router forwards a packet by examining the value of one or more fields in the arriving packet's header, and then using these header values to index into its forwarding table.
- The value stored in the forwarding table entry for those values indicates the outgoing link interface at that router to which that packet is to be forwarded.

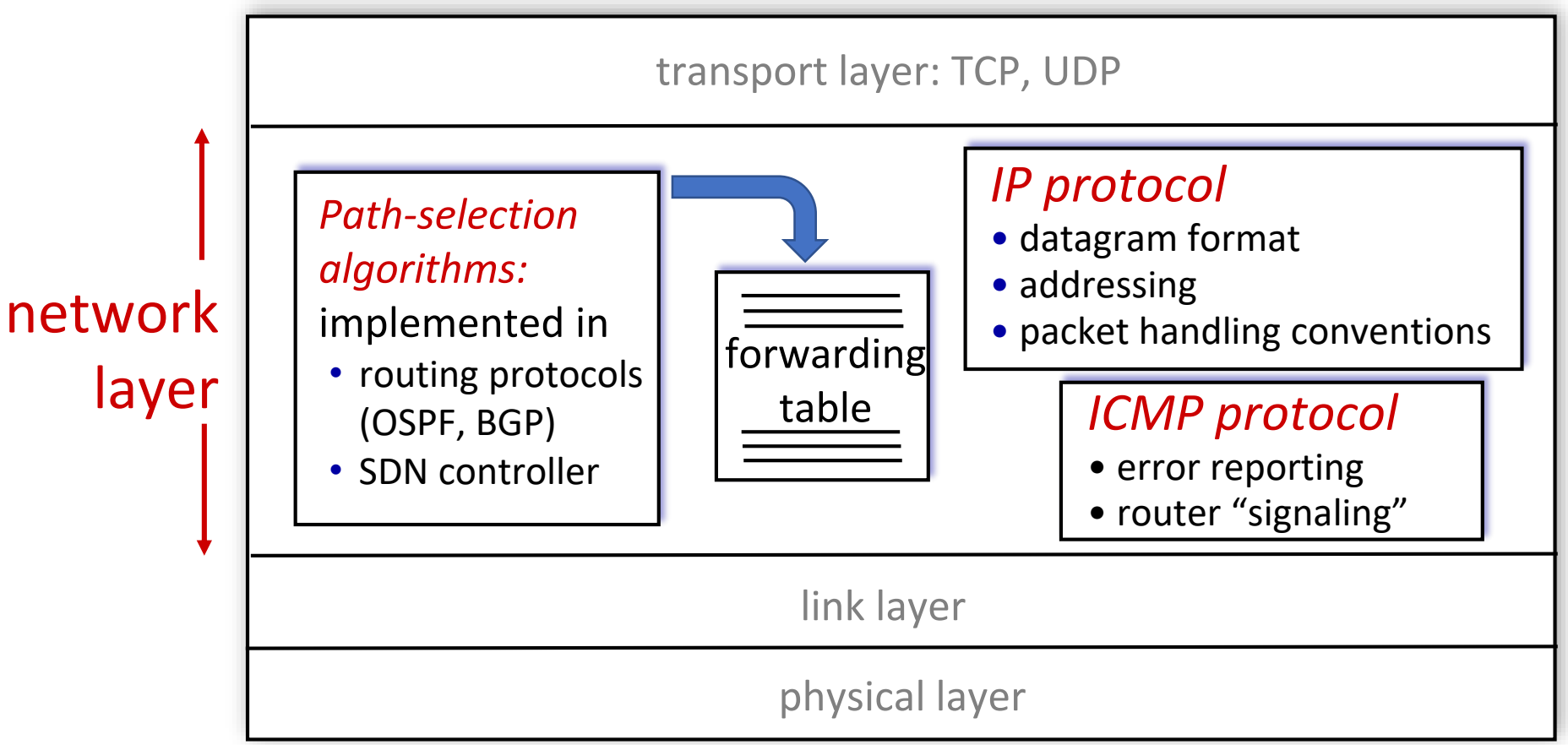
Network layer: “data plane” roadmap

- Network layer: overview
  - data plane
  - control plane
- What’s inside a router
  - input ports, switching, output ports
  - buffer management, scheduling
- IP: the Internet Protocol
  - datagram format
  - addressing
  - network address translation
  - IPv6

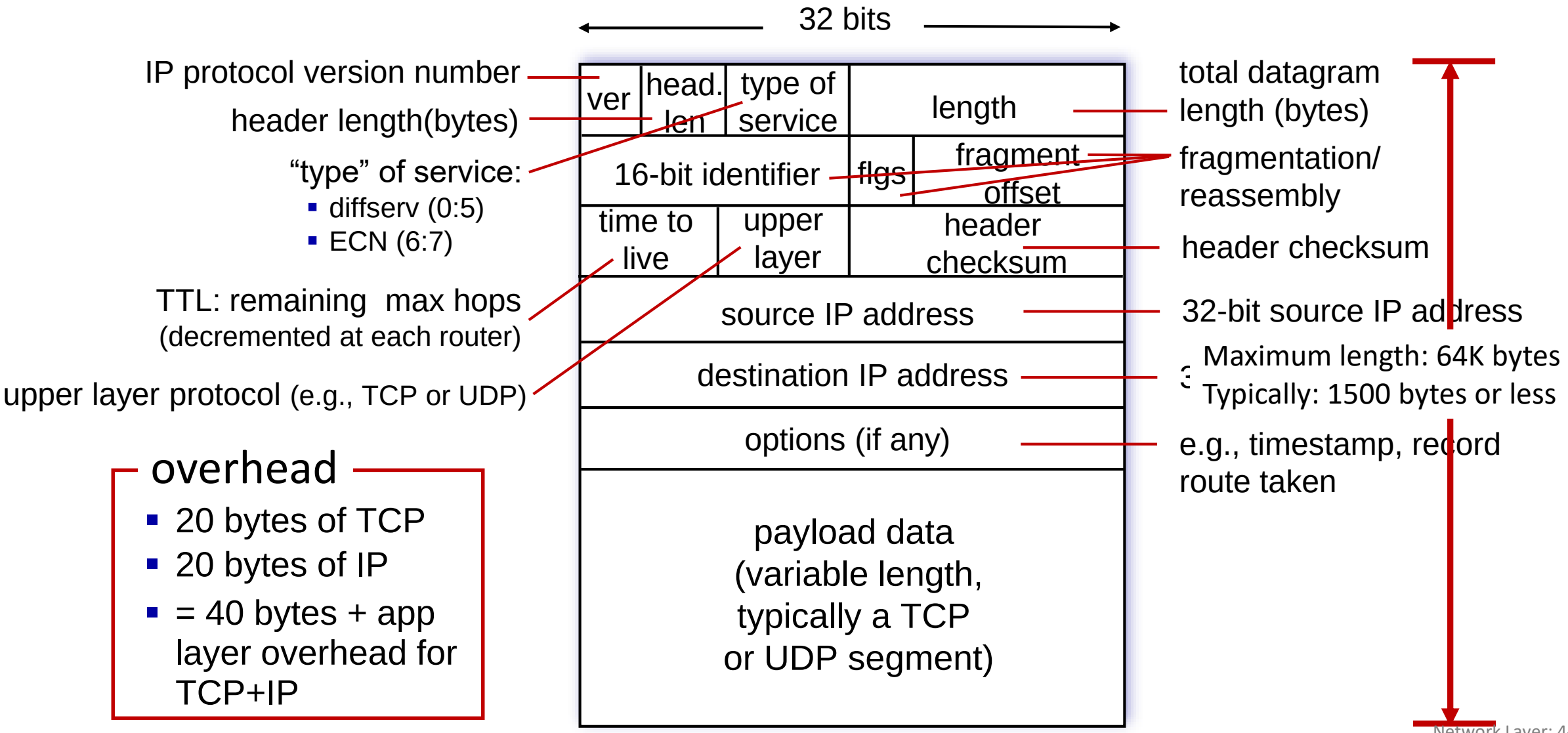
- Generalized Forwarding, SDN
  - match+action
  - OpenFlow: match+action in action
- Middleboxes

Network Layer: Internet

host, router network layer functions:



IP Datagram format

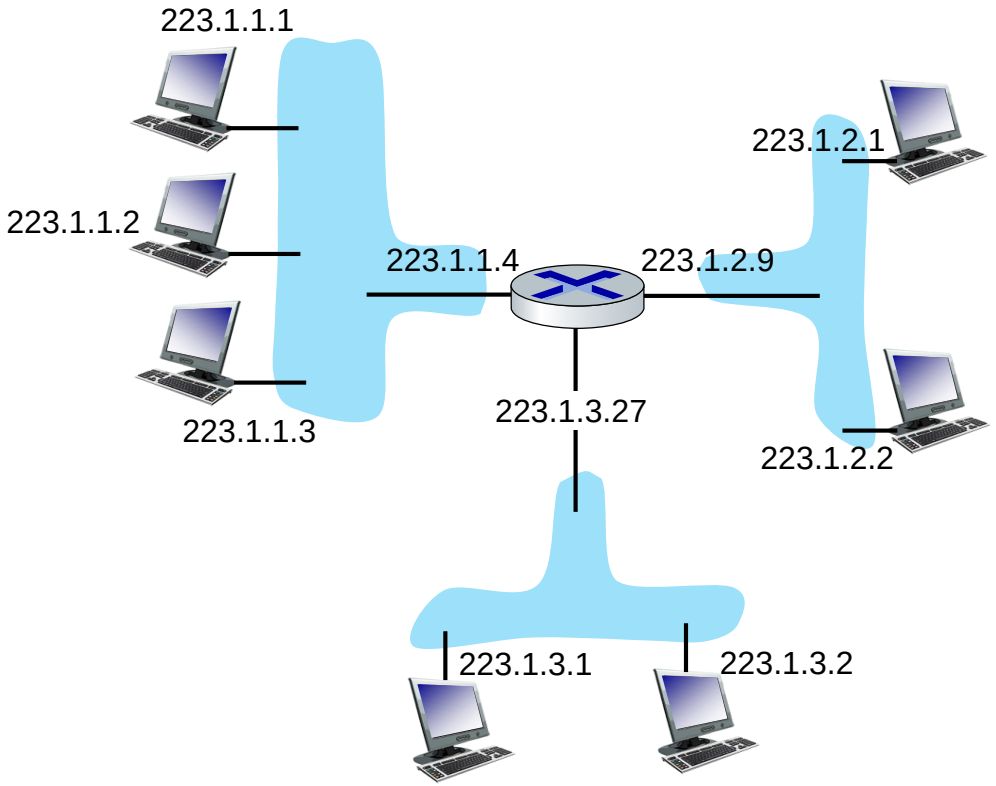


overhead

- 20 bytes of TCP
- 20 bytes of IP
- = 40 bytes + app layer overhead for TCP+IP

IP addressing: introduction

- IP address: 32-bit identifier associated with each host or router *interface*
- interface: connection between host/router and physical link
  - router's typically have multiple interfaces
  - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

IP addressing: last words ...

Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers http://www.icann.org/

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , …) management

Q: are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

"Who the hell knew how much address space we needed?" Vint Cerf (reflecting on decision to make IPv4 address 32 bits long)

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